

CMSC 207: Chapter 8 Homework

May 3, 2026

1. Let R be a relation defined on the set of all integers $\mathbb{Z}$ as follows:

$$\text{for all integers } m \text{ and } n, m R n \iff 5 \mid (m - n).$$

Prove that R is an equivalence relation on $\mathbb{Z}$.

Let m , n , and p be arbitrary integers.

Proof: R is reflexive.

Since $5 \mid (m - m)$ is equivalent to the statement $5 \mid 0$, which is true, $m R m$, so R is reflexive.

Proof: R is symmetric.

Suppose $m R n$. By the definition of R , we have $5 \mid (m - n)$. This means that there exists some integer k such that $m - n = 5k$. By elementary algebra:

$$m - n = 5k \rightarrow n - m = 5(-k).$$

Since $-k$ is an integer, by the definition of divisibility, $5 \mid (n - m)$. By the definition of R , we have $n R m$. Therefore, R is symmetric.

Proof: R is transitive.

Suppose $m R n$ and $n R p$. By the definition of R , we have $5 \mid (m - n)$ and $5 \mid (n - p)$. This means that there exist some integers k_1 and k_2 such that $m - n = 5k_1$ and $n - p = 5k_2$. By elementary algebra:

$$m - n = 5k_1 \rightarrow m = n + 5k_1.$$

$$n - p = 5k_2 \rightarrow n = p + 5k_2.$$

Substituting the second equation into the first gives:

$$m = (p + 5k_2) + 5k_1 = p + 5(k_1 + k_2).$$

$$m - p = 5(k_1 + k_2).$$

Since $k_1 + k_2$ is an integer, by the definition of divisibility, $5 \mid (m - p)$. By the definition of R , we have $m R p$. Therefore, R is transitive.

Since R is reflexive, symmetric, and transitive, R is an equivalence relation on $\mathbb{Z}$. ■

2. Let S be the set of all strings of 0s and 1s of length 3. Define a relation R on S as follows:

for all strings s and $t \in S$, $s R t \iff$ the two left-most characters of s and t are the same.

Prove that R is an equivalence relation on S .

S can be expanded as follows:

$$S = \{000, 001, 010, 011, 100, 101, 110, 111\}.$$

Let s , t , and u be arbitrary strings in S .

Proof: R is reflexive.

Since the two left-most characters of s are the same, $s R s$, so R is reflexive.

Proof: R is symmetric.

Suppose $s R t$. By the definition of R , the two left-most characters of s and t are the same. This means that the two left-most characters of t and s are the same, since the equality of strings exhibit an equivalence relation. By the definition of R , $t R s$. Therefore, R is symmetric.

Proof: R is transitive.

Suppose $s R t$ and $t R u$. By the definition of R , the two left-most characters of s and t are the same, and the two left-most characters of t and u are the same. This means that the two left-most characters of s and u are the same, since the equality of strings exhibit an equivalence relation. By the definition of R , $s R u$. Therefore, R is transitive.

Since R is reflexive, symmetric, and transitive, R is an equivalence relation on S . ■